Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/21/2012
PI Name:	Ianzini, Fiorenza Ph.D.		
Project Title:	Mechanisms of Cell Survival Following Space Radiation-Induced Mitotic Catastrophe: Implications for Cancer Risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	52242-1009	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	12/01/2010	**End Date:**	05/30/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date change to 5/30/2012 per NSSC information (Ed., 12/22/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mackey, Michael (University of Iowa)		
Grant/Contract No.:	NNX11AB60G		
Performance Goal No.:			
Performance Goal Text:			
	During space flights astronauts are exposed to various types of radiations from galactic cosmic rays (GCR) and solar particle events (SPE). A great concern of NASA's is the cytotoxic and genotoxic effects of these radiations while astronauts are circulating in geosynchronous orbit, are on mission at the International Space Station, or are commissioned to distant flights and space explorations. There are great uncertainties in the model of cancer risk used by NASA where radiation dependent cancer mortality rate is estimated from human epidemiological data for low LET radiations. The proposed studies specifically address the Research Emphases of this Solicitation and focus on characterizing novel molecular mechanisms thought to be involved in radiation carcinogenesis. As it pertains to carcinogenic processes, we will advance the understanding of radiation quality effects by using different types of radiations having different energies and fluencies and we will focus our		

Task Description:	studies on the role played by altered cell proliferation, aberrant mitoses, and aneuploidy/polyploidy cell formation. In line with these focuses, the principal goal of the present application is to test the hypothesis that cells undergoing mitotic catastrophe (MC-- these cells are large multinucleated, polyploid cells) after exposure to charged particle radiation are able to escape death, to undergo depolyploidization, to give rise to a cell progeny of small mononucleated cells that regain a degree of genomic stability sufficient for them to survive, to proliferate, to acquire transformation competence, and thus to become a focal spot for cancer initiation. Thus, these studies will determine the fate of cells subjected to charged particle radiation and will shed light on the carcinogenic risk from radiation exposure. [Ed. Note: Grant continues as "Mechanisms of Cell Survival Following Space Radiation-Induced Mitotic Catastrophe: Implications for Cancer Risk--NNX12AG50G"--see that task for future reports.]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have participated into three NSRL campaigns from December 2010 (when this grant started) to May 2012. During these irradiation shifts we have utilized proton beams and silicon and iron ions of different energies and LET, and both human normal epithelial cells and fibroblasts. We are still analyzing the data for protons and silicon ions while we have completed analysis for iron ions with the MRC-5 cells (lung fibroblasts). One of the most important findings of our studies is that we were able to demonstrate that all the cell lines we have analyzed undergo space radiation-induced MC and that they are able to survive for weeks post-irradiation. These results, coupled with findings on cancer cells (Ianzini et al. 2009) that demonstrated that irradiated cells undergo MC, become polyploid, and are able to escape death by activating processes of depolyploidization that utilize uncharacteristic pathways of cell divisions (namely activation of meiotic genes during multipolar divisions), prompted us to hypothesize that similar processes might occur in normal cells exposed to space radiation. Our studies, indeed demonstrate that this is the case. We have found that MRC-5 cells not only undergo iron ions-induced MC and escape death, but also become polyploid, and express both meiosis-specific genes and specific markers of cancer stem at times matching the depolyploidization processes as detected through live cell imaging (Kosmacek et at. 2012). Iron ions-irradiation also induce persistent and elevated phosphorylation of H2AX (?-H2AX); furthermore, iron ions irradiated polyploid cells are not senescent (Kosmacek et at. 2012). The fact that gamma-H2AX is persistent and statistically significantly elevated is important because we find that gamma-H2AX positive cells are all those with large and/or multinucleated nuclei. These findings provide support for the hypothesis that the polyploid cells possess enhanced enzymatic activity where proteins involved in the nucleosomal organization/maintenance of chromatin (such is H2AX; Hacques et al. 1990; Lindsey et al. 1991; Usachenko et al. 1994; Zheng et al. 2003) are stimulated, possibly to facilitate an overall genomic repair and/or reorganization better suited to sustain the cells' proliferation. Taken together, these data establish that iron ions irradiated polyploid cells maintain proliferative capability and acquire a cancer genotype (expression of meiosis-specific gene products and of cancer stem cell markers). These facts have great implications for carcinogenesis as the progeny so established are likely to carry a deficient or imperfect genetic composition that might render them prone to cell transformation. References Hacques M.F., Muller S., De Murcia G., Van Regenmortel M.H., Marion C. 1990. Use of an immobilized enzyme and specific antibodies to analyse the accessibility and role of histone tails in chromatin structure. Biochem. Biophys. Res. Commun., 168, 637-643. Ianzini F., Kosmacek E.A., Nelson E.S., Napoli E., Erenpreisa J., Kalejs M., Mackey M.A. 2009. Activation of meiosis-specific genes is associated with depolyploidization of human tumor cells following radiation-induced mitotic catastrophe. Cancer Res., 69, 2296-2304. Kosmacek E.A, Napoli E., Mackey M.A., Ianzini F. 2012. Iron ion radiation induces changes in ploidy, persistent activation of DNA repair enzymes and expression of the meiotic-specific protein SYCP3 in normal human fibroblasts. Radiation Research submitted. Lindsey G.G., Orgeig S., Thompson P., Davies N., Maeder D.L. 1991. Extended C-terminal tail of wheat histone H2A interacts with DNA of the ‘linker’ region. J. Mol. Biol., 218, 805-813. Usachenko S.I., Bavykin S.G., Gavin I.M., Bradbury E.M. 1994. Rearrangement of the histone H2A C-terminal domain in the nucleosome. Proc. Natl. Acad. Sci. USA., 91, 6845-6849. Zheng C. and Hayes J.J. 2003. Structures and interactions of the core histone tail domain. Biopolymers, 68, 539-546.
Bibliography Type:	Description: (Last Updated: 02/12/2021)
Abstracts for Journals and Proceedings	Kosmacek EA, Mackey MA, Colwell EA, Gridley DS, Ritter L, Ianzini F. "Human Epithelial Cells Exposed to 1GeV of Iron Ion Radiation Acquire Tumorigenic Potential and Stem Cell-Like Properties." 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. , Sep-2011
Abstracts for Journals and Proceedings	Kosmacek EA, Mackey MA, Ianzini F. "NEW VIEWS: DNA Demethylation May Play a Role in Space Radiation-Induced Cell Polyploidization." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012

Abstracts for Journals and Proceedings	Kosmacek EA, Mackey MA, Ianzini F. "Iron Ion Irradiation Produces Changes in DNA Ploidy of Human Normal Fibroblast Cells Without Affecting Proliferation or Cell Death." 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Articles in Other Journals or Periodicals	Kosmacek EA, Napoli E, Mackey MA, Ianzini F. "Iron Ion Radiation Induces Changes in Ploidy, Persistent Activation of DNA Repair Enzymes and Expression of the Meiotic-Specific Protein SYCP3 in Normal Human Fibroblasts." Radiation Research, submitted, as of August 2012. , Aug-2012
Articles in Peer-reviewed Journals	Meng X, Laidler LL, Kosmacek EA, Yang S, Xiong Z, Zhu D, Wang X, Dai D, Zhang Y, Wang X, Brachova P, Albitar L, Liu D, Ianzini F, Mackey MA, Leslie KK. "Induction of mitotic cell death by overriding G2/M checkpoint in endometrial cancer cells with non-functional p53." Gynecol Oncol. 2013 Mar;128(3):461-9. Epub 2012 Nov 9. https://doi.org/10.1016/j.ygyno.2012.11.004 ; PMID: 23146687; PMCID: PMC3612022 [Note previously reported in August 2012 as "PLosOne, submitted, as of August 2012."] , Mar-2013